6 The curve C has equation $y = \frac{x^2 + a}{x + a}$, where a is a positive constant.

(a) Find the equations of the asymptotes of C. [3]

(b) Find, in terms of a, the x-coordinates of the stationary points on C. [3]

(c) Sketch C, stating the coordinates of any intersections with the axes. [3]

(d) Sketch the curve with equation $y = \left| \frac{x^2 + a}{x + a} \right|$. [1]

(e) Find the set of values of a for which $\left| \frac{x^2 + a}{x + a} \right| = a$ has two real solutions. [4]